

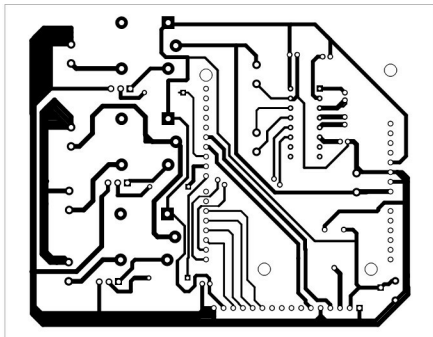


Fig. 2: Actual-size, PCB pattern of the SMS home automation system using Arduino

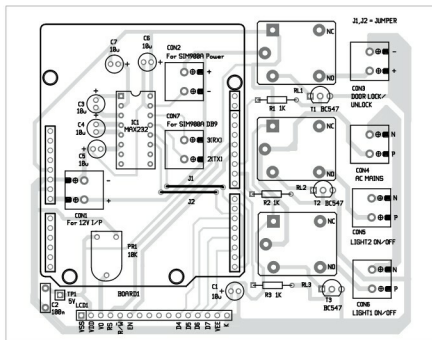


Fig. 3: Component layout of the PCB

on the LCD.

Send *LIGHT1OFF* as an SMS; light will turn off and corresponding string *LIGHT 1 OFF* will be displayed on the LCD.

Send *LIGHT2ON* as an SMS; light connected at CON5 will turn on and corresponding string LIGHT 2 ON

will be displayed on the LCD.

Sending *LIGHT2OFF* as an SMS again; light will switch off and corresponding string *LIGHT 2 OFF* will be displayed on the LCD.

Construction and testing

An actual-size, single-side PCB pattern of the SMS home automation GSM system is shown in Fig. 2 and its component layout in Fig. 3. For convenience, we have designed the PCB as an Arduino shield. You can modify the design as per requirement. Also, you can test the PCB with Arduino board using a cable connector.

Download the program in Arduino. Connect CON2 and CON7 with the GSM modem power and DB9 connector, respectively. Insert a SIM card in the GSM modem SIM slot. Power on the PCB using 12V supply at CON1 and AC mains at CON4 for electrical appliances.

Send an SMS to the inserted SIM card number to control the equipment. LCD1 as well as Arduino serial port will display the status message of the equipment.

Conclusion

The home automation market is growing very fast and is in need of a vast range of developments that can be carried out with a smarthome. For this project design and implementation of a smart GSM house was considered.

Arduino along with GSM provides smart automated home system with the desired baud rate of 9600bps. This prototype was implemented and tested with a maximum of three loads.

Use freewheeling diode (1N4007) externally across the relays RL1 through RL3 to prevent the damage to transistors from back emf. **EFY**



Biswajit Das was manager - R&D, EFY Labs till recently

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